

A LOW-COST LARGE-SIZE DOCUMENT SCANNER

Mandar Thite developed a low-cost scanner priced ten-times lower than scanners currently available in the market



Mandar Thite demonstrating his photo clipping machine

■ JALAJA RAMANUNNI

Design engineer Mandar Thite had many old newspapers stocked in his house as his father loved collecting newspaper clippings. Some of these were even over thirty years old. Thite wanted to immortalise these clippings by creating digital copies. However, he found

for a digital camera using a magnetic solenoid, motors and a rubberised roller. The apparatus includes a digital camera, a stand to hold it and a mechanism to feed documents to the camera continuously, so that these can be photographed. The digital scanner consumes only 15 watts of power. This innovation was recognised as one of the top five winners of 'The Next50

that scanning such large format papers would cost him a hefty sum of Rs 100 per sheet.

Thite decided to experiment with his 7.2-megapixel Canon Powershot A550 digital camera. He found that if he could digitally clip the captured images, it could turn out much cheaper. He developed a simple mechanism using easily available components like a camera, magnetic push-pull solenoid, permanent magnet DC motor and cycle tube along with a time synchronisation technique.

The scanner is created using a retrofit attachment

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How it works

In existing digital scanners images are captured by a charge-coupled device (CCD) array. A lamp is used to illuminate the document and the image of the document reaches the array through a series of mirrors, filters and lenses. The exact components vary according to the scanner models but the components are pretty much the same.

In Thite's digital scanner, a camera is mounted over a platform and a magnetic push-pull solenoid is fixed next to it. With power supply the solenoid gets pushed towards the camera's click button. The camera clicks a photograph of the document and instead of the lamp the flash provides the light. The image gets stored on its memory card. Though the solenoid and motorised roller are two separate units, these are synchronised to work together. These have separate controls with suitable 12V DC power supply. The maximum print-size depends on the area the camera can capture.

On the lower platform, the mechanism used is similar to that of a printer. Paper is fed into the machine by using a roller feeding system. The feed mechanism consists of an electrical motor for which power is required. A rubberised roller—in this case a PVC pipe wrapped with a simple cycle wheel's tube—is placed at the top of the papers. It is con-